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US Semiconductors and Semiconductor Equipment

Citi TMT Day 2: TXN, AVT, MPWR, INFQ, TER, MKSI, FORM, OLED Takeaways

CITI'S TAKE

We hosted 1) fireside chats with OLED, INFQ & FORM; 2) met TXN, MPWR, AVT, TER, and MKSI; and 3) hosted EDA and Quantum Panels on Day 2 of Citi Global TMT conference. See key takeaways inside.

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^{AC}See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations.

Citi TMT Conference Day 2

Texas Instruments (TXN)

- **Demand signals improving** – Lead times are beginning to extend and backlog trends continue to strengthen, while customers are showing greater urgency around securing supply. TI also highlighted improved visibility into customer ordering patterns and growth trajectories through better internal analytics.
- **Industrial recovery broadening** – A&D remains the strongest end market, including space-related programs, while industrial automation and medical are recovering. Consumer appliances remain the weakest sub-segment, although all industrial verticals grew YoY. TXN believes there has been some inventory replenishment at their end customers but channel inventory remains relatively lean revenue is still tracking below trend.
- **Secular content expansion intact** – Long-term industrial content growth continues to be driven by labor scarcity, energy efficiency requirements, and increased medical technology adoption.
- **Data center becoming a major growth vector** – Management sees a high likelihood that data center revenue ultimately exceeds the Personal Electronics segment. Incremental TAM is primarily concentrated in analog and power, particularly Stage 2 power, 800V architectures, and general-purpose regulators.
- **Stage 2 power is the next near-term opportunity** – Most current data center revenue comes from TI's catalog power portfolio, while Stage 2 power represents the largest incremental ASSP opportunity ahead.
- **Three drivers of data center growth** – TI sees growth coming from (1) AI infrastructure GW deployments, (2) 800V-driven content expansion, and (3) share gains as the company historically had limited participation in Stage 2 power.
- **800V adoption remains a long-term transition** – Management noted the industry required roughly a decade to transition from 12V to 48V architectures. The move toward 800V architecture will likely unfold over multiple years, though accelerated by NVDA and hyperscalers.
- **TI's GaN strategy**– TI believes its internal GaN manufacturing in Dallas and Aizu provides supply, scale, and cost advantages, while its GaN-on-silicon roadmap pushes beyond the industry's traditional sub-650V toward 900V-class applications in the data-center .
- **No interest in SiC** – TI does not participate in SiC FETs and views the market as increasingly commoditized, preferring to invest in integrated power-management solutions and high-voltage GaN where it sees greater differentiation and long-term returns. TI does not view that it will hinder their growth. SiC is typically used in SSTs.
- **China competition manageable** – TXN acknowledged Chinese competitors offer high-quality products and iterate rapidly, but Chinese suppliers are strongest in fragmented, niche markets, while TI believes its advantages remain portfolio breadth, scale, and manufacturing capacity. Management highlighted share gains in China during 2025 despite increased local competition.
- **Automotive remains structurally stronger than industrial** – Automotive avoided the 40% inventory correction seen in industrial markets due to ongoing

content expansion from safety mandates, body electronics, lighting, ADAS, and vehicle electrification.

- **Internal manufacturing remains a competitive advantage** – TI continues migrating production across 65nm, 45nm, and 28nm nodes into internal fabs, driving substantial cost savings versus external foundry sourcing.
- **300mm economics require scale** – Management argued that 300mm fabs require significant revenue scale (~\$5B), concentrated process technologies, and utilization to justify the investment, differentiating TI's business model from peers.
- **Capex remains disciplined** – Existing cleanroom capacity is sufficient for the next several years, with tool additions occurring incrementally alongside demand. Gross capex is expected to grow at roughly 1.2x the long-term revenue growth rate.

Avnet (AVT)

- **Lead times extending across the supply chain** – Lead times are broadly moving higher, with Samsung, SK Hynix, and Micron increasingly requiring LTAs, including through channel partners. Avnet is passing through higher memory costs to customers.
- **Memory becoming a larger revenue contributor** – Memory has increased to the mid-teens percentage of revenue from low-single digits several quarters ago, largely reflecting pricing inflation.
- **Pricing environment remains constructive** – Suppliers continue implementing price increases due to input-cost inflation, with most increases ranging from 10–25%. Management sees little evidence that any suppliers are reducing prices.
- **Inventory remains healthy** – Avnet inventory and customer inventory levels are generally in good shape, with backlog growth primarily driven by larger customers.
- **Smaller customers remain cautious** – Mass-market customers remain reluctant to build inventory following the last correction and continue ordering primarily against lead times rather than future demand.
- **Farnell provides an early demand signal** – Farnell's broad inventory position gives management visibility into the design funnel and serves as a leading indicator for future demand trends.
- **Auto customers increasingly value supply security** – Automotive OEMs remain focused on ensuring supply following COVID-era shortages and continue building buffer inventory strategies.
- **Chinese automotive suppliers outperform on supply-chain execution** – Management believes Chinese automotive suppliers generally demonstrate stronger supply-chain management capabilities than many Western peers.
- **Regional demand improving** – Growth accelerated across regions, including Americas (~16%), EMEA (~14%), and Asia (~12%), with Asia improving from roughly 5% the prior quarter.

- **Industrial, auto, and defense remain key end markets** – Industrial, automotive, and defense/aerospace remain the largest application areas, with defense/aerospace demand growing approximately 75% YoY.

Monolithic Power Systems (MPWR)

- **Enterprise Data outlook remains robust** – MPWR reiterated approximately 130% YoY Enterprise Data growth for 2026 and explained that the raised expectations in 2Q26 was driven by exceptionally strong 2Q bookings and book-to-bill.
- **No evidence of AI order pull-forward** – Management stated orders remain broad-based and does not believe customers are pulling in demand. Channel inventory remains lean.
- **Backlog visibility improving** – Backlog has expanded from roughly one quarter historically to approximately 1.5-2 quarters today, with some orders extending into 3Q27.
- **Data center share gains support growth above TAM** – Management believes MPWR can outgrow underlying AI infrastructure deployments through market-share gains, particularly among mid-sized and long-tail customers.
- **Customer additions accelerating** – Enterprise Data customer count has increased significantly since mid-2025, with many new opportunities shifting toward module-based architectures carrying 2-4x higher ASPs than discrete solutions. MPWR is present across all major GPU/ASIC shipments today.
- **NVIDIA remains on discrete solutions** – NVIDIA platforms remain largely on discrete power solutions, and MPWR believes migration on to modules could begin likely in Feynmann.
- **800V deployments remain a 2028 story** – MPWR continues to expect meaningful 800V deployments beginning in late 2027 and ramping through 2028, including incremental sidecar opportunities.
- **SiC remains the focus; GaN investment ongoing** – Management noted its current Stage 0 offerings remain SiC-based. MPWR currently incorporates third-party GaN technology alongside its controllers and is actively investing resources to develop its own GaN capabilities and offerings.
- **Content growth** – Power content should increase alongside higher TDPs, though management does not expect content to scale 1:1 with power consumption. Larger gains are expected from discrete-to-module transitions.
- **Management pushed back on share-loss concerns** – MPWR argued qualification cycles remain lengthy, and customers are unlikely to qualify numerous suppliers, and highlighted its time-to-market advantages.
- **Gross margin model unchanged** – Management maintains its long-term gross margin framework of approximately 55.5%.
- **Capital returns becoming more opportunistic** – MPWR indicated it becomes progressively more aggressive with share repurchases during periods of stock-price weakness.

- **CPO/NPO opportunity could exceed pluggables** – Management believes NPO and CPO opportunities could ultimately become larger than the traditional pluggable optics market, with CPO representing incremental content expansion.
- **Storage and computing remain sluggish** – Management expects storage and computing to remain roughly flat in 2026, while notebook demand remains down YoY.

EDA panel: Vera Yang – President & Co-Founder Flexcompute, Faraj Aalaei – Founder & CEO, Cognichip

- **Trust remains the moat** – Both panelists argued EDA incumbency remains durable because customers have massive sunk investment in trusted verification, sign-off, and IP workflows. Even if AI-generated designs improve rapidly, building confidence in autonomous design systems will take years given the cost and risk of tape-outs.
- **Nearest-term risk concentrated in Ansys-style simulation** – The most tangible area of AI-driven disruption discussed was multi-physics simulation for advanced packaging, warpage, thermal analysis, and photonics. This appears more directly competitive with the legacy Ansys portfolio than with core digital and analog EDA design tools.
- **AI enables cross-disciplinary design optimization** – Panelists highlighted that chiplets, HBM, optical interconnects, packaging, and thermal constraints increasingly require multi-physics analysis. Their AI approach aims to replace sequential engineering hand-offs with a unified workflow spanning multiple engineering domains.
- **Three key AI advantages** – The value proposition centers on (1) multi-physics modeling across engineering disciplines, (2) multi-scale simulation spanning nanometer-level detail to full-system designs, and (3) autonomous agents capable of running hundreds of design and validation iterations without human intervention.
- **Autonomous photonics design demo** – An AI agent reportedly reviewed technical literature, generated an initial photonics design, and completed roughly 200 optimization iterations in two hours, significantly reducing optical loss. Management indicated the platform is already deployed at approximately 8 of the 10 largest Chinese semiconductor companies.
- **Engineers remain in the loop near term** – Management emphasized customers are unlikely to trust unchecked AI-generated outputs for production tape-outs. Their adoption strategy therefore focuses on AI-assisted workflows that can be validated inside existing EDA environments, allowing trust to build incrementally.
- **Verification remains strategically important** – Even as AI expands design automation, panelists expect verification, validation, and sign-off functions to remain critical bottlenecks. Flexcompute noted their company can already originate photonics designs outside traditional EDA flows but still requires verification infrastructure before production deployment.
- **Consumption model could be a tailwind** – AI dramatically reduces simulation runtime while increasing total simulation activity. Instead of replacing EDA consumption, autonomous agents could drive substantially more tool usage

through continuous overnight and weekend optimization cycles, supporting consumption-based licensing models.

- **Simulation economics shift toward volume** – Examples cited simulation runtimes declining from hours to minutes, enabling engineers and autonomous agents to perform significantly more iterations per project. Faster simulations increase demand for compute capacity and software consumption rather than eliminating it.
- **10–20x customer ROI proposition** – Management framed customer value around faster development cycles, improved chip performance, and lower design risk. One example suggested reducing a project from three tape-outs to one could save roughly \$70M of development expense.
- **Large semiconductor R&D opportunity** – The panel claimed approximately 87% of semiconductor R&D labor is concentrated in device-level design activities, creating a sizable opportunity for AI-driven productivity gains.
- **Proprietary data remains a competitive barrier** – Unlike software, semiconductor design data is highly confidential and generally unavailable for foundation-model training. Management argued meaningful AI models require internally generated datasets derived from domain expertise and customer-specific workflows.
- **Engineering scarcity supports adoption** – Rather than replacing engineers, AI was positioned as a way to reallocate scarce talent toward architecture, system design, and innovation while automated agents handle repetitive simulation and optimization tasks.
- **AI could lower semiconductor startup barriers** – Rising design complexity and development costs have contributed to a sharp decline in new semiconductor company formation. Management argued AI-driven cost compression could make chip development economically feasible for a broader set of entrants.
- **Trusted validation becomes the bottleneck** – While startups may adopt AI-native design methodologies more quickly than incumbents, panelists argued long-term differentiation will increasingly depend on trusted verification and validation infrastructure rather than access to talent or capital.
- **Potential foundry implications** – AI-driven calibration of simulation results against historical fabrication data could reduce dependence on physical test runs and improve the correlation between design simulations and manufacturing outcomes.
- **Advanced packaging use case highlighted** – In one example, traditional tools could identify package failure but could not isolate the specific failing solder joint among hundreds of candidates. AI-based multi-physics models were presented as capable of predicting individual joint failures before fabrication.
- **AI may accelerate process/IP co-design** – AI could allow foundries, IP vendors, and chip designers to develop process technologies and compatible designs simultaneously rather than sequentially, potentially improving node adoption and foundry ROI.
- **Moat is data, workflows, and trust** – Management identified proprietary semiconductor datasets, recruited domain experts, and customer-specific engineering workflows as important competitive advantages. However, both

speakers ultimately emphasized customer trust as the most difficult moat to replicate in a highly confidential, mission-critical industry.

Infleqtion – INFQ

We hosted Infleqtion's CEO Matt Kinsella and recap key takeaways.

Neutral Atom Flexibility and Competitive Positioning: Management emphasized the high level of flexibility in the neutral atom quantum modality, uniquely positioning it to address both quantum sensing and quantum computing. Neutral atom's advantages vs other quantum approaches include room-temperature operation, lower capital intensity, and a simpler bill of materials. With regards to quantum computing, INFQ has significantly improved its qubit quality, and its systems are benefitting from the scalability of neutral atom qubits. Although INFQ expects industry consolidation ahead, they believe the market will ultimately support multiple quantum modalities, with the platform to first achieve useful quantum computing likely to gain share of customers, talent, and capital.

Solid revenue generation: Unlike many of its peers, INFQ is generating meaningful and growing revenues (~\$45M this year) from quantum sensors, quantum computing systems, and software offerings. Management highlighted its strategy of building a revenue-generating sensing business while continuing to invest in long-term quantum computing development.

100 Logical Qubit Key Target: Logical qubits, or error corrected physical qubits, are the most meaningful performance metric as they support useful computation. The industry will begin reaching quantum advantage at roughly 100 logical qubits, where quantum computers will be able to perform calculations that have commercial value. INFQ has achieved 12 logical qubits and is on track to reach 30 by the end of 2026. The company is targeting 100 logical qubits by the end of 2028.

Quantum and AI Relationship: Management emphasized that quantum computing is not a replacement for AI or classical computing, but a complementary technology capable of solving classes of problems that classical systems cannot efficiently address. Early applications are expected in molecular simulation, drug discovery, materials science, battery development, and other physics-based optimization problems. The company pointed to prior work with NVDA, where classical GPUs and quantum computing were used together on scientific modeling workloads, as a likely blueprint for future deployments.

NVDA Partnership and Ecosystem Development: INFQ characterized NVDA as a key enabler whose interests align with the advancement of quantum computing. NVDA contributes through their software tools, including Ising models, and NVQ Link, their hybrid quantum-classical infrastructure, as well as their willingness to work with other quantum providers.

Focus on R&D and scaling: The company's investment priorities remain split between advancing quantum computing performance and scaling quantum sensing products. Quantum computing R&D focuses on increasing qubit counts and improving fidelity, while sensing investments focus on reducing size and cost while increasing manufacturing scale. INFQ currently builds all of its technology internally but is evaluating the use of third-parties for final assembly as the business scales.

Quantum panel: IQM Quantum Computers – Jan Goetz (CEO & Co-Founder), Infleqtion – Matthew Kinsella (CEO)

Quantum will be key to solving hard problems – While panelists were mixed on the timing of the quantum computing journey (early days vs 2028 for reaching a key milestone) they were in agreement that there are complex problems that classic computers cannot solve that will require quantum computing. Importantly, quantum computers will be working with classical GPUs, not replacing them or AI.

US Government catching up – European governments have been investing in quantum for some time. Significant investments have been made in other countries as well, including the UK, Japan, China and Australia, and reflect a global recognition of the importance of quantum. With its recent \$2B CHIPS investment and White House executive orders, the US government is recognizing quantum's importance.

Teradyne – TER

Michelle Turner – CFO

Roy Chorev – VP & GM Compute Test Division

Kaleigh Ledbetter – Corp Comm & IR

- **AI Compute Test Intensity** — Management sees advanced packaging as the biggest structural driver of compute test TAM. As GPUs/AI accelerators add more compute die, HBM stacks, I/O chipleths and interconnects, each component must be tested individually and again at package level, while higher power, data rates and thermal requirements also increase test time and test-cell complexity. Teradyne therefore sees transistor/content growth as incremental to test demand than headline GPU unit growth.
- **GPU / Accelerator Intensity** — Large GPUs typically require materially higher test intensity than CPUs because package size limits parallelism, with about 4:1 ratio. Large AI accelerators can approach GPU-like intensity, while smaller inference accelerators sit somewhere in between. As examples, moving from two to four compute die can effectively double wafer-test requirements even before accounting for higher unit volumes. Existing testers can generally be reused across generations with incremental instrument/resource upgrades rather than full platform replacement.
- **Compute Share Gains** — Teradyne expects to gain roughly ~2 pts of compute share this year and aims to significantly narrow the gap with competitors. Share gains should come from both hyperscalers where Teradyne already has strong positions and penetration of accounts is historically stronger for competitors. Management emphasized that recent share mix was largely mix/customer exposure rather than losing any incumbent sockets, while Teradyne is now breaking into previously competitor-dominated accounts.
- **Second Hyperscaler** — Teradyne has now completed vendor qualification at a 2nd hyperscaler and is actively supporting its next-generation AI accelerator ramping in CY27. The device uses a complex supply chain in which the hyperscaler develops its own compute die while two design houses—one US-based and one Taiwan-based—participate in the packaged solution. Teradyne has already been selected by the hyperscaler for compute-die testing and has also made progress with one of the design houses on final/package test. Management emphasized that additional sockets remain to be decided across wafer and package test and can be controlled separately by the hyperscaler,

design houses or foundry, so the current wins represent meaningful initial penetration rather than the entire test opportunity.

- **Lead time/capacity** — Management views maintaining ~16-week lead times as a competitive advantage, particularly when customers already have Teradyne qualified as a second source or in memory where test programs are easier to port. Teradyne has 2x manufacturing capacity over the past six months and is investing heavily in inventory to preserve responsiveness. Memory is currently the biggest component constraint, with Teradyne buying ahead and paying premiums to secure roughly the next nine months of requirements. Management emphasized that manufacturing floor capacity alone is not the constraint; inventory, installation resources and field support must scale in parallel. The ability to ship and install faster can convert directly into incremental share if/when competitors are constrained.
- **Visibility** — Test visibility naturally comes later than WFE because Teradyne sits near the end of the semiconductor manufacturing chain. Customers must first secure wafer, HBM and advanced-packaging capacity before finalizing tester requirements, so firm forecasts may arrive only ~16 weeks before need. However, Teradyne has visibility through active NPI programs roughly six months before silicon, giving management directional insight into future ramps even before firm tester orders are placed.

MKSI Instruments (MKSI):

Michelle McCarthy - SVP & Chief Accounting Officer
Dave Henry - EVP Global Strategic Marketing
Pareto Misra - VP IR

- **SPE Outlook** — Management sees much better visibility this cycle, with major OEMs providing ~6–8 quarters of demand visibility and MKS passing 24–36 months forecasts to key suppliers. Customers are extending purchase windows, and MKS still expects semi revenue to significantly outgrow WFE, led by deposition/etch content.
- **SPE Capacity** — MKS believes its current footprint can support a ~\$250B WFE environment once the new Malaysia facility is fully built out, with temporary surge capability of ~20% that could support ~\$300B WFE for several years. Malaysia currently has one product line and should reach ~7–8 lines by YE27. Beyond that, MKS could expand its existing Mexico campus through a brownfield project in roughly ~18 months, materially faster than the Malaysia greenfield build.
- **SPE Mix** — Relative WFE exposure is lower than last cycle because China is largely unavailable and NAND has fallen from >20% of WFE to likely <10%. Even so, MKS expects strong WFE outgrowth, particularly in deposition/etch.
- **NAND Greenfield** — MKS sees visibility into NAND greenfield projects expected to come online around CY28, though these remain customer forecasts rather than firm orders. Management is expecting some upside bias that at least one customer could pull spending forward, potentially creating MKS demand around mid-CY27. Greenfield NAND is roughly 2x the MKS content of an upgrade because it includes vacuum, flow, valves, plasma and power, while upgrades are primarily high-power content.
- **Flow / Power** — MKS remains focused on higher-value integrated flow solutions rather than standalone MFCs. NAND is especially important for power because

high-aspect-ratio etch requires >100kW-class power, while DRAM and logic require much less. DRAM is beginning to add more high-aspect-ratio etch, but intensity remains well below NAND.

- **Pricing** — MKS maintains a value-based pricing strategy and reviews pricing continuously, but recent increases have largely offset tariffs, inflation and higher input costs rather than materially expanding margins. New products generally enter at higher ASPs/margins, while management is more measured on aggressively repricing legacy products given long-term customer relationships. Pricing should therefore improve structurally but is not being used simply to exploit the current supply-demand environment.
- **E&P** — Electronics & Packaging remains strong, with AI infrastructure increasingly driving HDI and multilayer-board demand in addition to package substrates. AI-related chemistry revenue has risen from ~5–10% historically toward ~20% exiting CY26, and management expects this mix to continue increasing as installed equipment drives subsequent chemistry pull-through. More complex HDI boards, additional layers and finer line/space all increase MKS equipment and recurring chemistry content.
- **E&P Equipment Margins** — MKS has implemented significant pricing increases on ESI drilling equipment, with current tool prices materially above CY25 levels, helping narrow the margin gap versus the corporate average. However, much of the backlog was contracted well in advance, limiting near-term realization; management expects more of the pricing benefit to flow through in late CY27 and CY28. Plating equipment carries even longer ~18-month commitments, delaying repricing further.
- **Chemistry equipment** — Management views commodity vertical plating as relatively unattractive but sees an opportunity in G-Plate for next-generation high-end AI boards approaching ~5/5 μm line/space. The technology flows chemistry across the panel rather than dipping it vertically and is currently being evaluated in Japan. MKS's strategy is to intersect the AI roadmap at advanced geometries rather than compete broadly in commoditized plating.
- **Incremental Margins** — Management continues to frame consolidated incremental margins around ~50%, though near-term margins are being burdened by upfront capacity and labor investments. Malaysia and incremental labor each represent tens of basis points of near-term drag, with Malaysia expected to become a tailwind as utilization improves toward YE27. Higher volumes should therefore drive increasing leverage once current capacity investments are absorbed.
- **Capital Allocation** — Capex should remain within the ~3–5% of revenue range while debt reduction remains the near-term priority. Leverage should approach ~2x by year-end, after which management expects a more balanced allocation approach. The bar for M&A is now much higher, with future deals likely limited to tuck-ins or technology gaps that cannot be filled organically.

FormFactor (FORM):

Michael Slessor – CEO

- **Test Intensity / Advanced Packaging** — Management sees advanced packaging and chiplets as the core structural driver of test growth. More die per package increases the need to test each chiplet before assembly and again at final package level, raising known-good-die requirements and probe-card spend per

unit. The historical ~0.4% probe-card intensity metric is becoming less useful because semiconductor ASPs, particularly DRAM, are rising much faster than probe-card pricing.

- **Custom ASICs** — FORM sees CY27 as the start of a meaningful transition in custom AI ASICs from legacy wire-based probe cards toward advanced MEMS as power, speed and thermal requirements increase. The company is engaged with major hyperscalers, foundries and OSATs, has already generated initial revenue, and expects more qualifications to ramp in CY27. Management views ASICs as a key contributor toward the \$1.6B CY30 revenue target.
- **GPU Opportunity** — FORM has completed qualification and is shipping pilot-production probe cards for GPU testing at the leading foundry. Management now believes its initial <\$100M annual GPU TAM estimate is likely conservative and sees a path toward ~20–30% share on a next-generation platform if its uptime and test-cell productivity advantages hold. Revenue contribution should become more visible in CY27.
- **HBM** — HBM revenue grew ~60–70% y/y in 1H26, driven by bit growth, share gains at a second customer and richer HBM4 high-speed configurations at 11Gbps. FORM remains the incumbent with the HBM market leader and is in active discussions with the third major DRAM customer for CY27/28 as capacity expands. Management views HBM de-specing as broadly neutral, with lower stack heights offset by more HBM units and potentially higher networking content.
- **Photonics / CPO** — Photonics remains an important emerging growth vector, with CPO revenue expected to reach ~\$20M by the end of 3Q26 versus an initial \$10–20M full-year expectation. FORM is engaged broadly across CPO, NPO and pluggables and is now seeing Triton HVM adoption broaden across multiple foundries and fabless customers beyond the initial concentrated customer set. Management views current test bottlenecks as reinforcing its incumbency and learning advantage rather than indicating a larger-than-expected technical problem.
- **Gross Margins** — Management continues to target ~55% gross margin. Most improvement is expected from self-help—higher yields, shorter cycle times and lower COGS—rather than broad pricing. Value-based pricing remains relevant in differentiated areas such as HBM high-speed test.
- **Capacity** — Capacity is currently the primary constraint on growth. The Farmers Branch facility should begin shipping qualification units in 4Q26 and ramp through CY27 to roughly 2x current California probe-card capacity. Management believes the site gives enough flexibility to support reaching the \$1.6B target ahead of CY30 if demand is strong enough.

Universal Display (OLED):

Brian Millard – CFO & Treasurer

- **Confidence on new growth phase.** Memory shortages are impacting their business near term, but fundamentals are strong long term. New capacity is coming online.
- **Demand drivers.** They expect foldables as a growing trend in smartphones over the next coming years. A foldable is an incremental opportunity compared to a non-foldable, which needs more OLED materials. \$20bn of capital investments

in 4 facilities on IT side, which is only 5% penetrated. Automotive is a growing segment as well, only 1% penetrated in OLED. Smartphone is 65% penetrated in OLED displays, mid and low end models are starting to introduce OLED.

- **New materials.** On smartphone side, smartphones will be flattish next year, but if more are from foldable that will be a benefit. In IT business, more units continue as capacity comes online.
- **Other display technologies.** Mini-LED is a form of LCD, OLED has taken share from Mini-LED. Micro-LED is an emerging technology, and some folks have de-prioritized this space. OLED cost effectiveness is a key enabler and beneficiary over Micro-LED.
- **Phosphorescent blue adoption.** OLED's customers are taking the long-term view on roadmap and R&D. Given the memory environment, they haven't seen things change and their customers continue to innovate and adopt new technologies.
- **Dollar content on phosphorescent blue.** This dollar content is at a premium over red and green. However, the company won't price it such that it would compromise the adoption.
- **OLEDX.** OLEDX allows for more light to get extracted out of the display. This can enable off-angle viewing to be the same as on-angle viewing. This can also improve the energy efficiency on a display by up to 2x.
- **Gross Margins.** Guided to 74-76% gross margins this year. On materials side, margins should increase in 2H.
- **Capital allocation.** Investing in business is the most important. On capital returns, they have had a dividend for north of 10 years and have increased it annually. In the last year, they started buying back stock.
- **Operating margins.** Guided to 34-36% this year, they have been north of that the last few years.
- **Annual Guidance.** They have provided annual guidance for a few years now, it's harder to guide quarterly because OLED has 6 major customers which can fluctuate.

Companies Mentioned:

FormFactor Inc (FORM.O; US\$114.18; 1; 09 Sep 26; 16:00) | Infleqtion (INFQ.N; US\$13.26; 1H; 09 Sep 26; 16:00) | MKS Inc (MKSI.O; US\$268.05; 1; 09 Sep 26; 16:00) | Monolithic Power Systems (MPWR.O; US\$1203.75; 1; 09 Sep 26; 16:00) | Teradyne Inc (TER.O; US\$383.69; 1; 09 Sep 26; 16:00) | Texas Instruments Inc (TXN.O; US\$261.59; 1; 09 Sep 26; 16:00) | Universal Display (OLED.O; US\$79.91; 2; 09 Sep 26; 16:00)

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Appendix A-1

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